

Literature Review

Problem Area: Classification of Mobile User Behavior from Device Usage Patterns

Ram — USN: 23BTRCO044 — CSE (IoT)

1. Problem Statement

Modern smartphones generate rich behavioral telemetry — app usage time, screen-on time, battery drain, data consumption and the number of installed apps — that reflects how differently individuals use their devices. The selected problem is to analyze and classify mobile users into distinct behavior categories (light to extreme usage) based on these usage and demographic attributes, and to understand which features most strongly drive battery and data consumption. This has direct applications in user segmentation, targeted app optimization, personalized recommendations and device power management.

2. Dataset Identification and Collection

The uploaded documentation describes an “Expanded User Behavior Dataset” of 1,000 rows and 11 columns, capturing device model, operating system, app usage time, screen-on time, battery drain, number of installed apps, data usage, age, gender and a labeled user behavior class. This structure closely matches, and appears to be an expanded version of, the publicly available “Mobile Device Usage and User Behavior Dataset” on Kaggle (contributed by Vala Khorasani), which contains 700 samples across the same eleven fields and labels users into five behavior classes ranging from light to extreme usage.

2.1 Source and Access

- Primary source: Kaggle — “Mobile Device Usage and User Behavior Dataset” (kaggle.com/datasets/valakhorasani/mobile-device-usage-and-user-behavior-dataset).
- Format: CSV, single flat file, no missing values reported by prior users of the dataset.
- Size: 700 base records (11 columns); the expanded variant referenced in the assignment scales this to 1,000 records with the same schema.
- Note from the dataset authors: the data is generated to simulate diverse, realistic usage scenarios and is intended for machine-learning practice rather than as a citable primary research source.

2.2 Why This Dataset Is Suitable

- It already contains a labeled target (User Behavior Class), making it directly usable for supervised classification without additional labeling effort.
- It mixes continuous usage metrics (app usage time, screen-on time, battery drain, data usage) with categorical demographic fields (gender, OS, device model), supporting both feature-engineering and fairness/segmentation analysis.
- Its scale (700–1,000 rows, 11 columns) is well suited to coursework-level model training with classical ML algorithms (decision trees, random forests, SVM, k-NN) without requiring heavy compute.
- It aligns closely with the datasets used in the peer-reviewed and preprint literature reviewed in Section 3, allowing methods from those studies to be adapted directly.

2.3 Limitations to Note

- The data is simulated rather than collected from real device telemetry, so conclusions should be framed as illustrative of methodology rather than as claims about real-world populations.
- The meaning of each numeric behavior class (1–5) is not independently defined, so class interpretation must be inferred from the distribution of underlying features.

3. Literature Review

The reviewed literature falls into two related streams: (a) predicting or classifying smartphone usage/behavior patterns using contextual and app-level data, and (b) predicting battery consumption or discharge from usage patterns, since battery drain is one of the strongest behavioral indicators in the dataset. Table 1 summarizes eight representative studies.

Ref.	Study / Focus	Method	Key Finding	Gap Relative to This Problem
[1]	Sarker et al. (2020) — Context-aware smartphone app-usage prediction	PCA for dimensionality reduction combined with a decision-tree classifier ("ContextPCA")	Reducing high-dimensional context data before classification improved precision, recall, F-score and ROC compared with using raw context features directly.	Focuses only on which app is used next, not on broader behavioural segmentation such as usage-intensity classes.
[2]	Sarker et al. (2019) — Comparing ML classifiers for personalized smartphone usage prediction	Benchmarked ten classical classifiers (ZeroR, Naive Bayes, Decision Tree, Random Forest, SVM, k-NN, etc.) on individual phone-log data	No single classifier dominated across all users; tree-based ensemble methods were generally the most effective for individual-level context prediction.	Evaluates per-user personalized models rather than a single generalized model across a population, and does not address demographic factors such as age or gender.
[3]	Survey on app usage prediction and recommendation — Literature survey of smartphone data mining approaches	Review of prediction techniques including deep reinforcement learning (DeepAPP) and privacy-preference modelling	App usage is closely tied to spatiotemporal context and individual personality traits, motivating profile-based prediction approaches.	As a survey, it synthesizes prior work rather than proposing a new model for a labeled behavior-class dataset.
[4]	Problematic smartphone use prediction study (2021) — Predicting smartphone addiction / overuse levels	Random Forest and Multilayer Perceptron classifiers with ANOVA-based feature selection on large-scale survey data	Usage-pattern features could reliably distinguish problematic from normal users, and linear discriminant analysis supported early-stage prediction of overuse.	Relies on self-reported survey responses rather than logged device metrics such as screen-on time or battery drain.
[5]	Smartphone app usage prediction using points of interest — Sequential and location-based app usage modelling	Mobility-trace features combined with sequence-prediction algorithms (e.g., APPM)	App usage shows strong sequential and habitual structure that is often independent of a user's immediate physical context.	Targets next-app prediction rather than classifying overall usage intensity or behavior category.
[6]	Battery discharge-rate prediction using ML (2016) —	Multilayer Perceptron and Support Vector Machine regression	Non-linear regression models outperformed simple linear estimators	Data collected from a single hardware platform, limiting generalizability across the

	Predicting phone battery discharge under different usage states	models trained on standby, video-playback and browsing states	for predicting device-specific discharge rate.	varied device models present in modern usage datasets.
[7]	Deep-learning battery-life model (2024) — Predicting battery consumption from app and network behavior	Deep neural network trained on application usage time, screen-on time, network type and battery temperature	Deep-learning models captured non-linear interactions between usage variables and battery drain more effectively than classical regression baselines.	Requires richer sensor-level telemetry than is available in simpler aggregate usage datasets.
[8]	Real-time battery life prediction (2018) — Battery-life estimation from comprehensive real-time usage traces	Portfolio of machine-learning predictors combined with a concordance-index approach to handle missing sensor data	Modelling systematic data-missingness explicitly improved prediction reliability over predictors that ignore missing values.	Built for continuous time-series telemetry rather than the daily-aggregate records typical of behavior-class datasets.

Table 1: Summary of reviewed literature on smartphone usage and battery-behavior prediction.

3.1 Synthesis

Across the reviewed studies, two consistent themes emerge. First, tree-based and ensemble classifiers (Decision Tree, Random Forest) repeatedly perform competitively for classifying usage or behavior categories from tabular, log-derived features, which supports their use as baseline models for the User Behavior Class target in the identified dataset. Second, battery drain is treated in the literature as both a predictive target in its own right and as a strong proxy signal for overall usage intensity — reinforcing its inclusion as a key feature (rather than only a label-adjacent variable) in behavior classification. A recurring gap across the reviewed work is the limited use of simple, demographic-inclusive tabular datasets: most prior studies rely on either raw device sensor logs or survey instruments, whereas the identified dataset offers a pre-aggregated, labeled alternative that is comparatively easy to model but has been less directly studied in peer-reviewed literature. This positions the selected dataset as a useful, lightweight testbed for applying and comparing the classification approaches surveyed above.

4. References

- [1] Sarker, I. H. et al. “ContextPCA: Predicting Context-Aware Smartphone Apps Usage Based on Machine Learning Techniques.” *Symmetry*, 12(4), 2020.
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- [5] “Smartphone App Usage Prediction Using Points of Interest.” *arXiv:1711.09337*, 2017.
- [6] “Two Machine Learning Models for Mobile Phone Battery Discharge Rate Prediction Based on Usage Patterns.” *Journal of Information Processing Systems*, 2016.
- [7] “Enhancing Smartphone Battery Life: A Deep Learning Model Based on User-Specific Application and Network Behavior.” *Electronics (MDPI)*, 13(24), 2024.

- [8] "Predicting Smartphone Battery Life Based on Comprehensive and Real-Time Usage Data." arXiv:1801.04069, 2018.
- [9] Khorasani, V. "Mobile Device Usage and User Behavior Dataset." Kaggle, 2024. kaggle.com/datasets/valakhorasani/mobile-device-usage-and-user-behavior-dataset